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Research Article

Using COSMO-RS to Predict Solvatochromic Parameters for Deep Eutectic Solvents

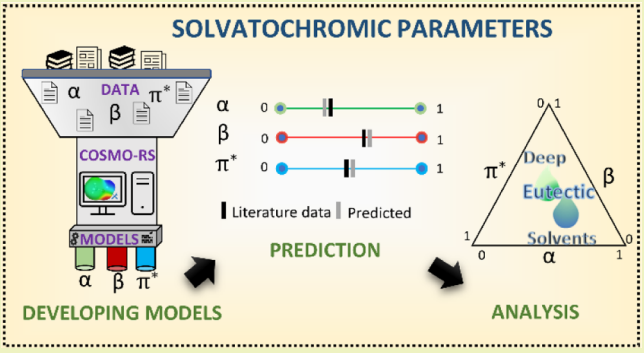
José Pedro Wojeicchowski, Dinis O. Abranches, Ana M. Ferreira, Marcos R. Mafra,* and João A.P. Coutinho*

Cite This: *ACS Sustainable Chem. Eng.* 2021, 9, 10240−10249

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ABSTRACT: The development of novel green solvents demands the knowledge of their properties, such as polarity, which can be described through solvatochromic parameters. However, while these are available for a wide range of conventional solvents, there is a lack of data for the emergent ones. Considering the need for such data, predictive models to estimate the Kamlet−Taft (K−T) parameters for deep eutectic solvents (DES) are developed here. The models, based on the conductor-like screening model for real solvents (COSMO-RS) descriptors, were initially developed and tested for 175 organic solvents to validate the applicability of the proposed approach. This approach was then extended for DES, which were classified into two categories, acids and nonacids. The developed equations showed a very good performance for all three K−T parameters, and this is the first work to propose models for all K−T parameters for DES. Moreover, a comparison between polarity data of DES and organic compounds showed that DES, rather than replace common solvents, can extend their range of polarities, reinforcing their designer solvent ability.



KEYWORDS: *deep eutectic solvents, Kamlet−Taft, polarity, acidity, basicity*

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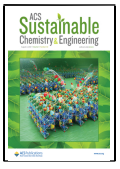
INTRODUCTION

Since the establishment of green chemistry and the increasingly stringent requirement for a cleaner chemical industry, the scientific community has been active in developing more sustainable chemical processes and products.[1] Part of this effort is dedicated to solvents, which represent around 80% of the total volume of industrial chemicals and continue to be a major source of environmental contamination due to their toxicity and volatility.[2] Several greener alternatives to classical volatile organic solvents have been developed over the years, namely solventless systems,[3] aqueous systems,[4] supercritical solvents,[5] ionic liquids,[6] and deep eutectic solvents (DES).[7]

DES are liquid systems prepared by physically mixing solid precursors, which are usually Lewis or Brønsted acids and bases that act as hydrogen bond donors (HBDs) or hydrogen bond acceptors (HBAs).[7] The liquid phase of DES arises not due to a chemical reaction (e.g., acid−base) between HBDs and HBAs but rather due to the formation of strong favorable interactions between HBAs and HBDs that enhance the melting point depression of a eutectic-type system when compared to systems that would present an ideal behavior.[8] Much like ionic liquids, DES are designer solvents because their properties can be tailored by the careful choice of HBAs and HBDs. As designer solvents, DES have been applied in different areas, namely, extraction of phenolic compounds,[9] metal ions,[10] and organic pollutants;[11] purification and concentration of solutes/analytes;[12] analytical chemistry;[13] supported liquid membranes,[14] and synthesis applications.[15] Furthermore, many DES are easily prepared, cheap, biodegradable, and nontoxic, making them attractive candidates as green solvents.[16]

To develop novel industrial applications for green solvents, their properties, such as thermophysical properties, as well as their solvation properties, such as polarity, must be known.[17] Since the polarity of a solvent governs its interactions with itself and with other compounds, it is a key parameter in selecting a solvent. Solvatochromic parameters such as the Kamlet−Taft (K−T) are useful to describe the polarity of a solvent.[17] They are the dipolarity/polarizability (π^*),[18] which describes the polarizing ability, the basicity (β),[19] which describes HBA ability, and the acidity (α),[20] which describes hydrogen-bond donating ability. Each of these parameters is

Received: April 17, 2021
Revised: June 28, 2021
Published: July 16, 2021



ACS Publications

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scaled to two key reference solvents and most solvents will have values between 0 and 1 on each scale.[17] For instance, π^* could be referenced to cyclohexane for a value of 0, and dimethyl sulfoxide is given a value of 1.[18] Moreover, formic acid and ethylenediamine show high values of α and β, respectively.

Despite the usefulness of the K−T parameters, they are still not available for a wide range of unconventional solvents such as DES or ionic liquids. In fact, the complexity of these solvents hinders the measurement of their corresponding solvatochromic parameters. For example, the poor solubility of Reichardt's dye in aqueous solutions of ionic liquids prevents the accurate measurement of their hydrogen bond acidity. To overcome this issue, Freire and co-workers[21,22] developed an alternative probe based on NMR spectroscopic data. This novel technique has also been applied to highly acidic DES.[22]

Several computational tools have been developed to predict the solvatochromic parameters of ionic liquids.[23,24] Cláudio *et al.*[25] used the conductor-like screening model for real solvents (COSMO-RS) to estimate the hydrogen bond basicity (β) of ionic liquids and Kurnia *et al.*[24] extended their method to predict hydrogen bond acidity (α). The prediction of solvatochromic parameters of DES has so far been addressed by few authors. Recently, Sherwood *et al.*[26] attempted to use COSMO-RS to predict the K−T parameters of DES but found the method to overestimate the values of α. Another work proposing correlations for α and β parameters for DES, based on interaction energies provided by COSMO-RS, was presented by Kundu *et al.*[27] These are relevant and promising results but an in-depth study to develop an accurate predictive model for all K−T parameters remains a challenge.

Considering the need for predictive methodologies for the K−T parameters of DES, a predictive methodology based on COSMO-RS is developed in this work to estimate the K−T parameters (α, β, and π^*) of these alternative solvents. First, correlations are developed and tested for conventional organic solvents, which demonstrate the applicability of the methodology, then the methodology is extended to DES and, finally, their polarity is compared to that of organic solvents.

COMPUTATIONAL METHODS

Conductor-like Screening Model for Real Solvents. To use COSMO-RS, the geometry and screening charge density of each individual molecule must be known. This was obtained for all compounds studied in this work using the COSMO-BP-TZVP template of the TmoleX software package (TURBOMOLE V7.3 2018).[28,29] This template includes a def-TZVP basis set for all atoms, DFT with the B-P83 functional level of theory, and the COSMO solvation model (infinite permittivity). Ionic compounds were optimized using the electroneutral mixture approach, where ionic compounds are optimized as ion pairs rather than separate ions.[9,30,31]

All COSMO-RS calculations were performed using the software package COSMOthermX (Version 19.0.1)[32,33] with the BP_TZVP_19.ctd parametrization. Note that the choice of using the BP_TZVP parametrization, rather than BP_TZVP_Fine, was based on (i) the fact that the computational cost of using a def2-TZVPD basis set and a fine grid would be prohibitively high for such a large database as the one presented in this work, and (ii) both parametrizations have been shown in the literature to yield results with the same level of accuracy or, in some cases, BP_TZVP has been shown to be superior to BP_TZVP_FINE.[34−36]

For each compound, the following properties were calculated using COSMO-RS: (i) misfit interaction energy (E_{MF}); (ii) van der Waals interaction energy (E_{vdW}); (iii) moment HB acceptor (M_{HBA}), and (iv) moment HB donor (M_{HBD}). Only the set of descriptors for the pure components are needed to estimate the K−T parameters. This approach is especially important for DES. Since they are mixtures, their descriptors will be computed individually, not as a sum of ions as done by other authors working with ionic liquids, which are pure compounds. Each COSMO-RS descriptor of the DES is calculated as a composition weighted average of each COSMO-RS descriptor for the pure HBA and pure HBD, as will be shown. Furthermore, our approach can easily be extended to encompass ternary or multinary DES systems, such as DES/water mixtures, without a loss of generality.

K−T Parameters Modeling. Based on a data set of 175 pure solvents by Marcus[37] (Table S1, Supporting Information), a linear regression analysis was carried out in order to create correlations to estimate α, β, and π^* parameters. After preliminary studies with organic solvents, the same procedure was extended to a data set of 44 DES classified into 2 groups (acids and nonacids). The experimental data used, and their COSMO-RS parameters are reported in Tables S2 and S3 (Supporting Information), respectively. About 2/3 of each database was used to develop the models (training set), and the remaining data to their validation (testing set). The deviation of predicted K−T parameters with the experimental data was estimated by the mean squared error (MSE), mean absolute error (MAE), and coefficient of determination (R^2).

RESULTS AND DISCUSSION

K−T Parameter Model for Organic Solvents. The first step of this work is to develop correlations for the K−T parameters of conventional organic solvents based on the descriptors of COSMO-RS. This was done considering the K−T parameters of 175 organic solvents as reported in the compilation of Marcus,[37] which, to date, remains the most complete database of K−T parameters assessed under the same experimental conditions.[26,37] From this data set, 117 were used as the training set to develop the correlations and the remaining 58 as the testing set.

Molecules are considered as a group of surface segments within the COSMO-RS framework, and each segment possesses a particular screening charge. To calculate thermodynamic quantities, COSMO-RS relies on pairwise interactions between each segment pair. This framework is advantageous in this work because parameters can be defined that quantify specific interaction abilities of molecules. For instance, M_{HBA} and M_{HBD} quantify the ability of a molecule to behave as a HBA or a HBD, respectively, making them invaluable to construct correlations for KT parameters. From there, misfit energy originates from the nonideal contact of two segments and van der Waals energy is the result of dispersion energy plus cavity formation energy.[27] The COSMO-RS descriptors chosen to correlate the K−T parameters of the solvents were M_{HBA}, M_{HBD}, E_{vdW}, and E_{MF}. These parameters have been used in the development of predictive correlations numerous times in the past, such as in the prediction of the logarithmic soil sorption coefficient prediction (K_{OC})[38] or the octanol−water partition coefficient (log K_{OW}).[39]

The calculated COSMO-RS descriptors for each compound are available in Table S1, Supporting Information. Considering the nature of the descriptors M_{HBA}, M_{HBD}, and E_{vdW}, they were chosen to be used in the correlation of α, according to eq 1. Note that a complete list of evaluated equations is available inTable S4 (Supporting Information), where the models used in this work are highlighted.

$$\alpha = x_1 M_{HBA} + x_2 M_{HBD} + x_3 E_{vdW} + x_4 \quad (1)$$

The correlations obtained (fitted parameters reported in Table 1) were used to predict the α of the remaining 58

https://doi.org/10.1021/acssuschemeng.1c02621
ACS Sustainable Chem. Eng. 2021, 9, 10240−10249

Table 1. Adjusted Coefficients for α, β, and π^* Parameters for Organic Compounds

coefficients	α	β	π^*
x_1	9.652×10^{-3}	1.540×10^{-1}	-3.955×10^{-2}
x_2	2.563×10^{-1}	8.828×10^{-3}	7.818×10^{-2}
x_3	1.995×10^{-3}	3.080×10^{-2}	3.033×10^{-1}
x_4	2.141×10^{-2}	1.104×10^{-1}	-2.828×10^{-2}
R^2	0.86	0.63	0.42

compounds in the database (testing set), and the results are depicted in Figure 1. This figure demonstrates the capability of the correlation to predict the parameter α of organic solvents. It is worth highlighting that 53 of the 58 predicted α values were inside the 95% prediction interval. In addition, the deviations between experimental and predicted data, measured by the MSE and MAE, were equal to 0.037 and 0.088, respectively (Table S5, Supporting Information).

Given the success in correlating the α parameter, the COSMO-RS descriptors M_{HBA}, M_{HBD}, and E_{vdW} were further used to correlate the β parameter, using the same dependency previously adopted for α

$$\beta = x_1 M_{HBA} + x_2 M_{HBD} + x_3 E_{vdW} + x_4 \quad (2)$$

The predictions of the correlation for the training and testing sets are depicted in Figure 2. The results show that only a few points are outside the 95% confidence interval, even for the testing set, indicating a good predictive ability of the model. The variation of the predicted data evaluated by MSE and MAE was equal to 0.076 and 0.229, respectively, as reported in Table S5.

Having successfully developed linear models for α and β, this approach was then extended for π^*. Although the structure of eq 3 that will be used to correlate π^* is identical to those used for α and β, there is a difference in one of the COSMO-RS descriptors

$$\pi^* = x_1 M_{HBA} + x_2 M_{HBD} + x_3 E_{MF} + x_4 \quad (3)$$

After various preliminary attempts, it was found that the use of misfit energy instead of the van der Waals energy led to a better model. The π^* takes into account the nonspecific interaction such as dipole−dipole and dipole−induced dipole interaction. From the COSMO-RS point of view, E_{MF} describes the unbalance of charge density of two neighboring molecules whose surfaces are close. Considering that, a polar molecule present in the system can induce a dipole in a neutral molecule due to the net charge difference involved. Furthermore, the result of two dipolar molecules interacting with each other is the dipole−dipole interaction. Note that M_{HBD} and M_{HBA} keep their importance once they represent the charged areas of the molecule and therefore are more susceptible to polarization. However, E_{vdW} exhibits lower influence mainly because π^* does not describe specific interaction but overall polarizability, making E_{MF} rather suitable than E_{vdW} for modeling this parameter.

The correlation coefficients obtained are presented in Table 1, resulting in a R^2 value equal to 0.42, which although seems low, the results presented in Figure 3 show that few data are outside the confidence intervals, for both training and testing points.

The fitted parameters obtained for the three correlations described above are reported in Table 1. The results show that, as expected, the HBD moment has a higher impact on the correlation for α, while the HBA moment governs the results for β. This means that the developed models are more than just simple mathematical equations but are soundly based on the physical phenomena behind the solvatochromic parameters. The R^2 values for the correlations could be improved if the data were classified into different families of compounds, such as carboxylic acids, amines, ketones, alcohols, and so forth. However, the goal here was the development of a general model for each parameter that would have a good predictive ability using as few predictive parameters as possible.

The results reported above demonstrate that COSMO-RS descriptors can be successfully used to correlate the solvatochromic parameters and predict the polarity of organic solvents. It is expectable that this method could be also extended to the prediction of solvatochromic parameters for nonconventional solvents, such as DES, for which data are scarce and difficulties may be found in the experimental measurements, such as the physical state of the DES or interferences with the Reichardt's dye probe.[22]

K−T Parameter Modeling for DES. There are a few previous studies addressing the prediction of K−T parameters for DES. Sherwood applied some COSMO-RS descriptors to correlate with rate constants and equilibria and calculate the solvatochromic parameters.[26] Despite the results obtained, the authors cannot determine the hydrogen bond donating ability for DES. Concerning the other solvents evaluated, α was estimated using seven parameters, taking values from a specific range of the σ-profile, based on a work of Palomar.[40] Kundu *et al.*[27] developed correlations based on COSMO-RS to predict α and β parameters, but no investigation was made about π^*. Besides, in order to improve the global accuracy, they created

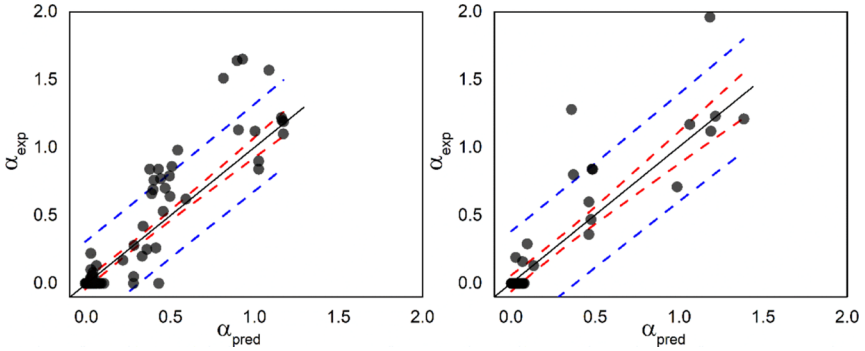


Figure 1. Experimental (α_{exp}) *vs* predicted (α_{pred}) values of the hydrogen bond acidity parameter, using correlations developed in this work, for the training (left) and testing (right) sets for organic compounds: dashed lines are the 95% prediction (blue) and confidence (red) intervals.

https://doi.org/10.1021/acssuschemeng.1c02621
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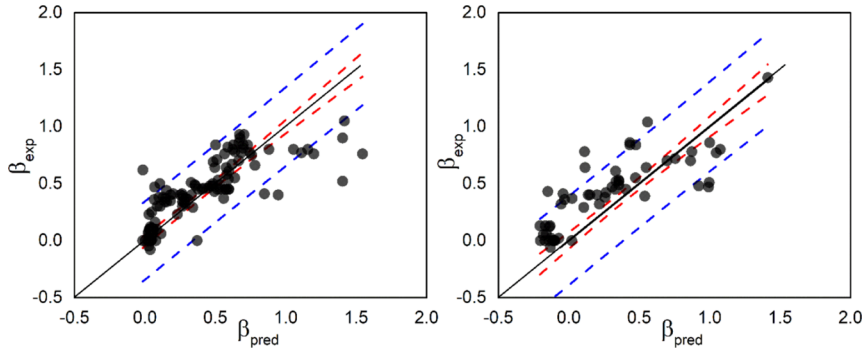


Figure 2. Experimental (β_{exp}) *vs* predicted (β_{pred}) values of the hydrogen bond acidity parameter, using correlations developed in this work, for the training (left) and testing (right) sets for organic compounds: dashed lines are the 95% prediction (blue) and confidence (red) intervals.

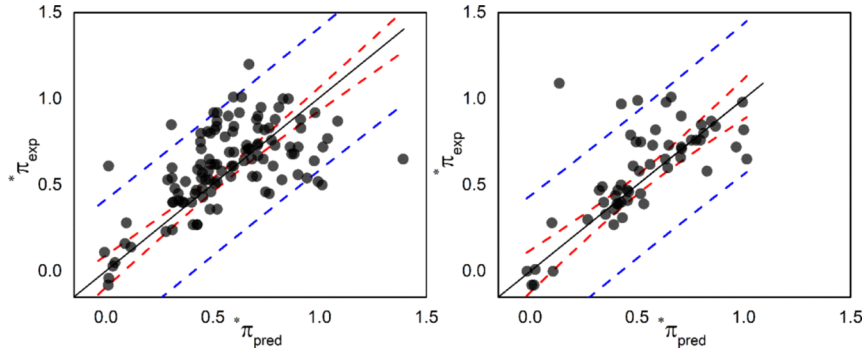


Figure 3. Experimental (π^*) *vs* predicted (π^*) values of the hydrogen bond acidity parameter, using correlations developed in this work, for the training (left) and testing (right) sets for organic compounds: dashed lines are the 95% prediction (blue) and confidence (red) intervals.

Table 2. Adjusted Coefficients for α, β, and π^* Parameters for Acid and Nonacid DES

	acids			nonacids		
coefficients	α	β	π^*	α	β	π^*
x_1	-8.414×10^{-2}	3.589×10^{-2}	2.737×10^{-2}	2.65×10^{-1}	-1.49×10^{-2}	2.02×10^{-2}
x_2	1.753×10^{-2}	-2.071×10^{-1}	9.214×10^{-2}	-6.95×10^{-2}	-7.06×10^{-2}	9.04×10^{-2}
x_3	2.371×10^{-2}	-4.494×10^{-2}	3.157×10^{-2}	-3.74×10^{-1}	-5.99×10^{-2}	9.43×10^{-2}
x_4	1.967×10	6.885×10^{-1}	1.784×10^{-1}	-3.62×10	5.83×10^{-1}	4.13×10^{-1}
R^2	0.797	0.816	0.831	0.867	0.707	0.823

various classes of DES, based on the HBA compound ([Ch]Cl, [N_{4444}]Br, [N_{4444}]Cl, menthol, and betaine). Moreover, for the DES formed by increasing the alkyl chain length of the HBD, keeping the same HBA, the authors proposed another set of equations to predict the solvatochromic parameters. Thus, considering all possibilities, eight models were applied for each K−T parameter estimation, showing the difficulties they faced in obtaining a generic model for this purpose.

Using a different approach, the ability of COSMO-RS descriptors to correlate the K−T parameters of classical solvents was demonstrated in this work, yielding simple, linear, yet accurate correlations. From now, the proposed method was used to describe α, β, and π^* of DES, based on literature data for 44 DES (Table S2, Supporting Information).[22,41−44] Note that the DES database selected presents a wide diversity, ranging from small, densely charged ions to largely hydrophobic components such as fatty acids.

Experimental data were grouped in two different DES categories: (i) acids (solvents that contain an acid HBD) and (ii) nonacids (solvents with other kinds of HBDs). For DES formed by acid HBDs, the correlations were developed using a training set with 18 experimental data points and a testing set with 9 DES. For DES composed of nonacid HBDs, the training set is composed of 11 data points and the testing set has6 experimental values.

The correlation models developed to predict α, β, and π^* parameters were based on the same descriptors provided by COSMO-RS in eqs 1−3. Since DES are not a pure compound, unlike ionic liquids or the compounds studied above, but instead a mixture of two or more components, the models to calculate the K−T basicity and acidity were different from those proposed by Cláudio *et al.*[25] and Kurnia *et al.*[24] While these authors took in account the sum of cation and anion ionic-liquid energy parameters from COSMO-RS to develop the equations to determine the K−T parameters, we considered COSMO-RS parameters individually from each HBA and HBD that constituted DES. In order to calculate the descriptors used in 4, a weighted average of the individual descriptors was calculated, taking into account the molar fraction of each component in the DES. The weighted descriptors are represented by a horizontal bar above their symbols, as shown below in eqs 4−6. The complete lists of evaluated models are available in Tables S6 and S7

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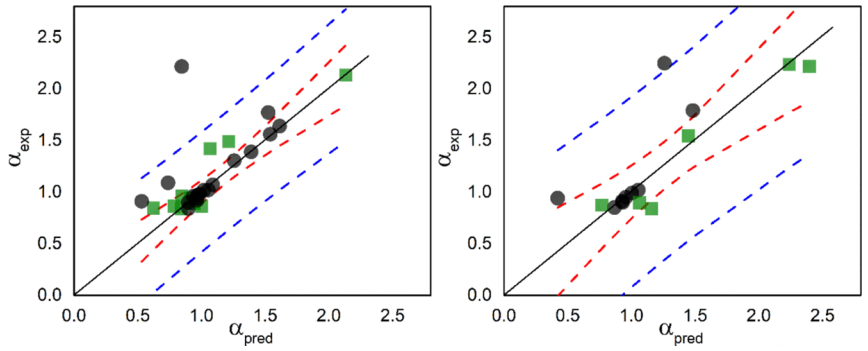


Figure 4. Experimental (α_{exp}) *vs* predicted (α_{pred}) values of the hydrogen bond acidity parameter, using correlations developed in this work, for the training (left) and testing (right) DES sets, according to the following groups: acid DES (gray, ●), nonacid DES (green, ■). Dashed lines are the 95% prediction (blue) and confidence (red) intervals.

Table 3. Experimental and Predicted Solvatochromic Parameters (α, β, and π^*) for the Different DES in the Study

model group	DES[a]	HBA[b]	HBD	α_{exp}	α_{pred}	β_{exp}	β_{pred}	π^*_{exp}	π^*_{pred}
acids	1	$[N_{2222}]Cl$	butanoic acid	0.990	1.004	0.760	0.779	0.920	0.857
	5	$[N_{3333}]Cl$	hexanoic acid	0.910	0.929	0.920	0.905	0.850	0.871
	8	$[N_{4444}]Cl$	hexanoic acid	0.900	0.931	1.020	0.953	0.810	0.862
	10	$[N_{4444}]Cl$	decanoic acid	0.850	0.867	1.280	1.075	0.690	0.870
	14	$[N_{4444}]Br$	butanoic acid	1.020	1.054	0.810	0.850	0.930	0.831
	16	$[N_{4444}]Br$	butanoic acid	0.940	0.421	0.820	1.573**	0.950	0.960
	19	$[N_{4444}]Br$	decanoic acid	0.950	0.956	1.050	1.038	0.710	0.842
	36	[Ch]Cl	citric acid	2.248	1.260**	0.714	−0.599**	1.373	1.364
	44	Menthol	dodecanoic acid	1.790	1.483	0.570	0.682	0.370	0.642
nonacids	20	[Ch]Cl	urea/glycerol	0.893	1.159	0.583	0.555	1.170	1.189
	24	[Ch]Cl	glycerol/ethylene glycol	0.897	1.065	0.625	0.603	1.127	1.145
	25	[Ch]Cl	ethylene glycol/formamide	0.872	0.769	0.657	0.601	1.113	1.106
	35	[Ch]Cl	D-sorbitol	2.216	2.398	0.493	0.470	1.244	1.533**
	37	[Ch]Cl	zinc chloride	1.543	1.446	1.253	0.933	1.406	0.810**
	38	betaine	glycerol	2.236	2.239	0.587	0.565	1.253	1.219
		R^2		0.926		0.861		0.891	
		MSE		0.019		0.013		0.011	
		MAE		0.096		0.073		0.071	

[a]Classification according to Table S2 (Supporting Information). [b]$[N_{2222}]Cl$: tetraethylammonium chloride; $[N_{3333}]Cl$: tetrapropylammonium chloride; $[N_{4444}]Cl$: tetrabuthylammonium chloride; $[N_{3333}]Br$: tetrapropylammonium bromide; $[N_{4444}]Br$: tetrabuthylammonium bromide; MSE; MAE; **not considered in statistical analysis.

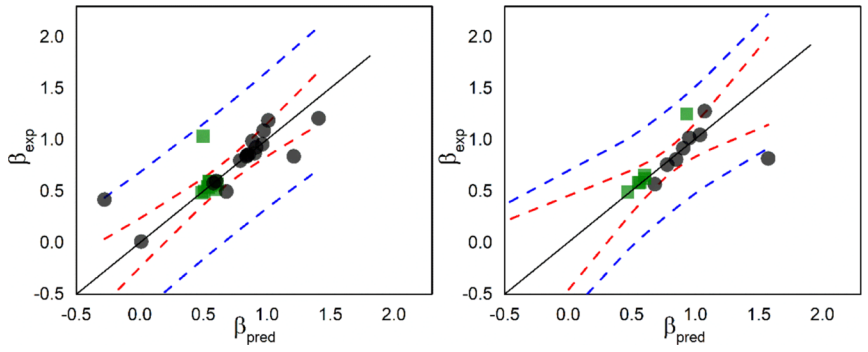


Figure 5. Experimental (β_{exp}) *vs* predicted (β_{pred}) values of the hydrogen bond acidity parameter, using correlations developed in this work, for the training (left) and testing (right) DES sets, according to the following groups: acid DES (gray, ●), nonacid DES (green, ■). Dashed lines are the 95% prediction (blue) and confidence (red) intervals.

(Supporting Information), where the models used in this work were highlighted.

The study was started by the correlation of the hydrogen bond acidity parameter (α) that was correlated by a linear equation described here as eq 4

$$\alpha = x_1\overline{M_{HBA}} + x_2\overline{M_{HBD}} + x_3\overline{E_{vdW}} + x_4 \quad (4)$$

All fitted coefficients reported in Table 2 support the physical meaning of the correlations proposed in this work. For instance, in the α correlation for acidic DES, x_1 (the coefficient associated with M_{HBA}) is negative, while x_2 (the coefficient

https://doi.org/10.1021/acssuschemeng.1c02621
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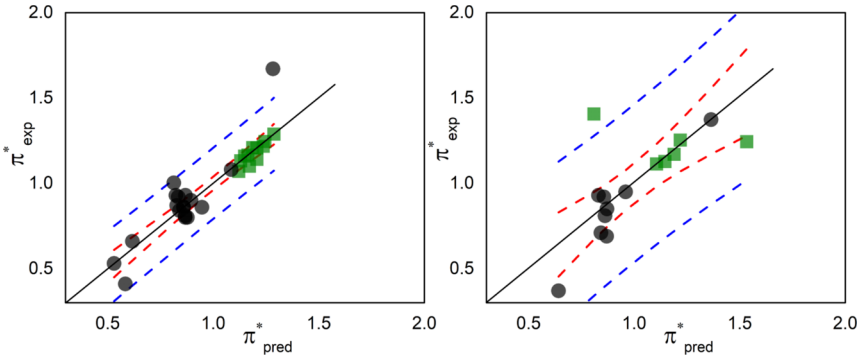


Figure 6. Experimental (π^*_{exp}) *vs* predicted (π^*_{pred}) values of the hydrogen bond acidity parameter, using correlations developed in this work, for the training (left) and testing (right) DES sets, according to the following groups: acid DES (gray, ●), nonacid DES (green, ■). Dashed lines are the 95% prediction (blue) and confidence (red) intervals.

associated to M_{HBD}) is positive. This explicitly shows that having a stronger HBA in a DES decreases its acidity perhaps due to the formation of strong molecular interactions that lock the HBD and decrease its ability to behave as a HBD toward other molecules. The same reasoning can be applied in the β correlation for acidic DES, as it will be shown, where x_1 is positive, while x_2 is negative. It is not clear why this behavior is not seen in the correlations for common organic compounds (previous section) or for nonacidic DES but could be related to the intermolecular interaction strength between DES components, which should be stronger in acidic DES. Furthermore, the higher complexity of the nonacid DES studied in this work prevents a full interpretation of these results.

The coefficients of the correlation for the α parameters for DES showed that the model presents a good description of the experimental data, with an R^2 value of 0.797 and 0.867, respectively, for the acid and nonacid DES. The graphical representation of experimental *versus* predicted data presented in Figure 4 gives a good description of the 15 experimental data points in the training set. Note that most of the points are within the confidence interval. Table 3 confirms this observation, with low MSE and MAE error values, associated with a high R^2 (0.926) between the experimental α and the predicted data for the testing set.

According to Table 2, the correlation used for the β parameter, eq 5, showed a slightly different performance than that for the α parameter and presents a good R^2 (>0.707).

$$\beta = x_1\overline{M_{HBA}} + x_2\overline{M_{HBD}} + x_3\overline{E_{vdW}} + x_4 \quad (5)$$

Figure 5 shows the correlation between the experimental and calculated β parameter for DES in the training (left) and testing (right) sets. A good description of the testing set is shown by the very few points outside the confidence interval and just one out of the prediction interval. Hence, after removing the two outliers, the errors evaluated by MSE and MAE were low, equal to 0.013 and 0.073, respectively, Table 3.

Unlike previous works[26,27] that proposed correlations only for two of the three solvatochromic parameters, this work pioneers the development of a correlation for all solvatochromic parameters for DES. An approach based on a linear dependency of COSMO-RS descriptors will be followed, as previously done for the calculation of α and β, to correlate π^* for DES, in the form of eq 6

$$\pi^* = x_1\overline{M_{HBA}} + x_2\overline{M_{HBD}} + x_3\overline{E_{MF}} + x_4 \quad (6)$$

The descriptors selected were the same previously used for the organic compounds discussed above, and as for the α and β, they are weighed by the concentration of the HBD and HBA. According to the results reported in Table 2, the correlation obtained from the training data set presented a high R^2 (>0.82), indicating a good correlation.

Figure 6 shows the relationship between the predicted and experimental values of the π^* parameter (Table 3) for 15 DES used to test the correlation. The good performance for the π^* parameter was reinforced by the low MSE and MAE values of 0.013 and 0.07, respectively. This demonstrates that the approach here proposed could be used to create a reliable correlation for the polarizability of DES.

Differently from other works, in our approach all three K−T parameters can be predicted successfully. Despite the good results obtained in this study, generally, Kundu *et al.*[27] found better accuracy predicting α and β, Table S8 (Supporting Information). However, the enhancement in the model's performance was attached to the higher number of equations applied, eight for each K−T parameter. It is worth highlighting that our method classifies DES into two groups (acids and nonacids), and only the descriptors of pure compounds are needed.

Moreover, probably those authors made a mistake modeling betaine-based DES by confusing the values of π^*_{exp} with those of α_{exp}, that is, the correct value of α_{exp} of betaine/glycerol was 2.236, not 1.253,[44] impacting the confidence on this specific model group. In its turn, Sherwood *et al.*[26] proposed a different way to obtain π^* and β, based on the linear correlation of K−T with the logarithmic of reaction equilibrium constant, which was calculated using the "reaction" function of COSMO-therm.[26] For α value calculation, they used descriptors from the σ-profiles of pure compounds but failed to calculate α for DES due to an overestimation of hydrogen bond donating ability. Finally, although the literature showed that σ-moments (such as HBD moment and HBA moment) did not correlate with K−T parameters,[26] this work proved the contrary.

Polarity of Organic Solvents *Versus* DES. Figure 7, based on a proposal by Dwamena and Raynie,[45] presents a ternary plot for the three normalized K−T parameters to compare and better understand the polarity of the 44 DES and 175 organic solvents used in this study. As some experimental α, β, and π^* values exceed the extreme limits (0−1), normalization was performed as per the following steps: (i) sum of the absolute values of the cells in the row was obtained;

https://doi.org/10.1021/acssuschemeng.1c02621
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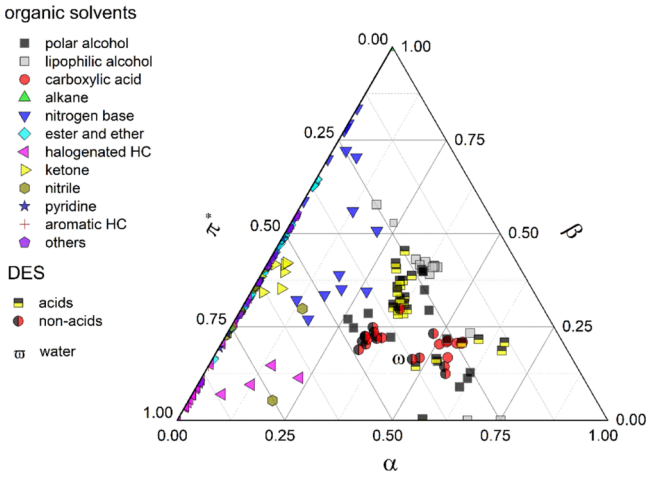


Figure 7. Plot of ternary experimental normalized K−T parameters of organic solvents, DES, and water.

(ii) each cell value of this row was divided by the sum calculated from the previous step.

Here, it is possible to notice that different DES groups exhibited their results in clusters, reinforcing the idea that performing regression analysis using two different categories, acids and nonacids, was a good insight.

From Figure 7, it is possible to observe that, generally, acid DES are within the range of lipophilic alcohols, and nonacid DES are close to the polar alcohols. Some of these DES, such as the menthol-based, are hydrophobic and have been used as tunable solvents to extract bioactive substances from plant biomass[46] as well as metal ions.[47] These are the yellow half squares furthest to the right of the graph, close to a specific nonpolar solvent, octanol. In general, the values for the α parameters of DES are higher than those for most organic solvents, revealing the greater hydrogen bond acidities of these solvents.

In the last decade, DES have been considered potential substitutes to conventional organic solvents, useful for several processes, especially for bioactive compound extraction.[44,48−50] As shown by Cai *et al.*[51] their hydrogen bond ability can be evaluated by the value of the $\alpha \times \beta$ of the solvents. They showed that the extraction of solutes increases with the increase of the value of $\alpha \times \beta$ since higher values lead to a more favorable interaction between the target solute and the solvent. Thus, the proposed correlations should help in solvent selection and design, saving time, costs, and reducing the number of laboratorial experiments. However, note that there

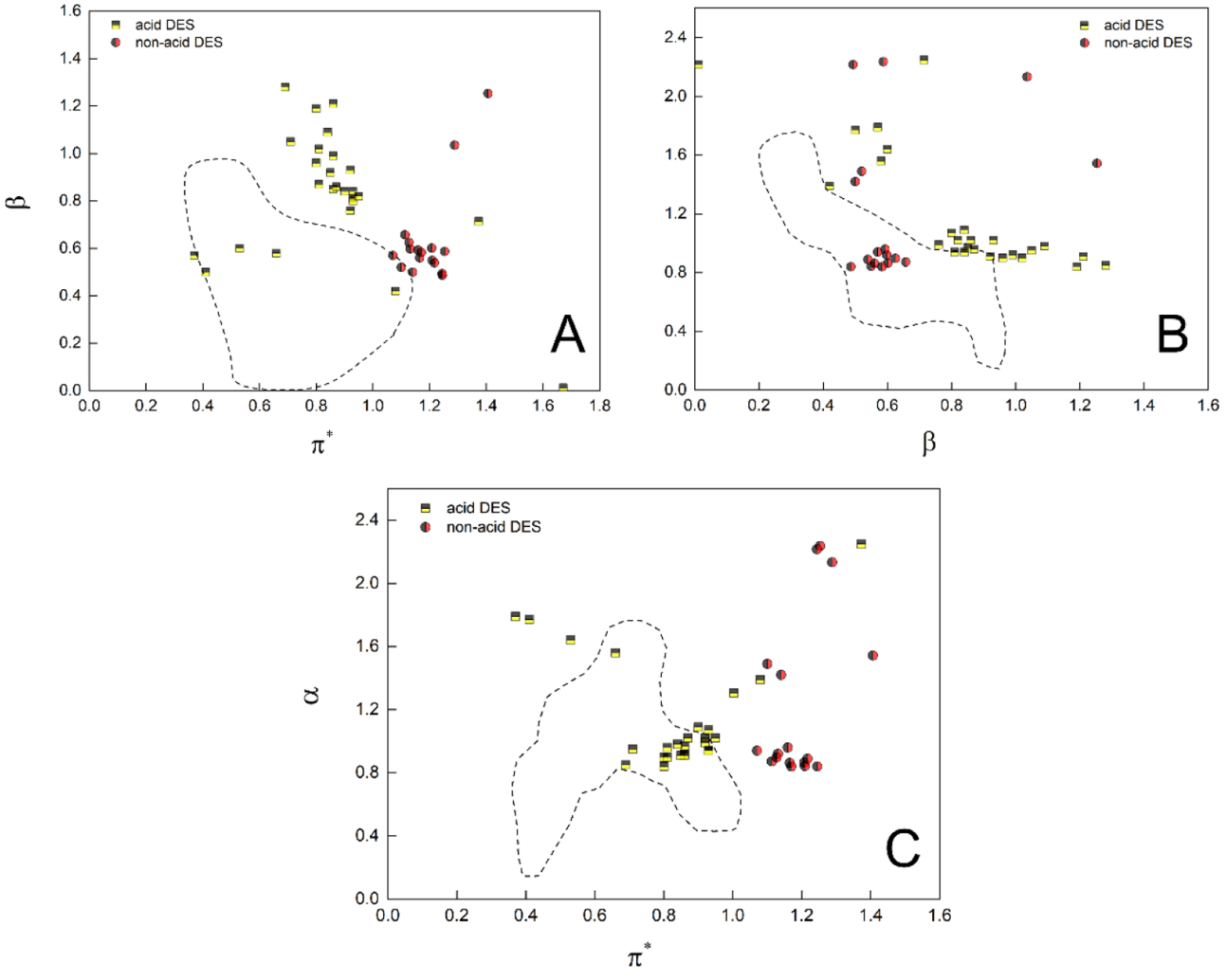


Figure 8. DES plotted according to their π^* *vs* β (A), α *vs* β (B), and α *vs* π^* (C) values. The area enclosed by the dashed line refers to the range of properties possessed by the set of Marcus database solvents.

https://doi.org/10.1021/acssuschemeng.1c02621
ACS Sustainable Chem. Eng. 2021, 9, 10240−10249

are other factors, besides hydrogen bond interaction, which can also influence the solvent selection, namely hydrophobic interactions, electrostatic interactions, and van der Waals forces.[51]

Jessop[2] pointed out that the first grand challenge of green chemistry is ensuring that green solvents are available to replace any kind of nongreen solvents. To understand better the possibilities of that happening, Jessop proposed a plot π^* *versus* β using the most common solvents, protic ($\alpha > 0.5$) and aprotic ones ($\alpha < 0.5$), and shaded the area populated by data points.[2] Through this approach, the plot was divided into different regions of polarity and basicity, allowing the comparison between these shaded areas for classical and green solvents. However, the alternative solvents did not fill completely the region occupied by the organics, and some processes require, for now, the use of solvents that are not green.

Following this approach, we evaluated the protic solvents based on the Marcus database for comparison with the DES, and the results for the common organic solvents are similar to those reported by Jessop[2]. When the results obtained for DES are plotted in Figure 8, where the dashed line corresponds to the zone of common organic solvents (Figure S1, Supporting Information), it is possible to see that rather than replacements of common organic solvents, DES are actually novel solvents that extend the range of polarities of available solvents, which reinforce their designer solvent character.

It is clear from Figure 8A that the nonionic DES studied here (menthol-based, yellow half squares in the left) overlap the region of common organic solvents but cover only a small part of their zone. The same was observed on α *versus* β and α *versus* π^*plots, Figure 8B, and 8C. Furthermore, in this study, all DES formed by acid HBDs were classified as "acid". However, as shown in Figure 8A, part of acid-DES presents β values corresponding to high basicity. Thus, it is clear that the hydrogen bond basicity of DES (above the dashed line, in the middle of the graph) is governed by the ammonium-based salt, that is, the HBA, as suggested by Teles *et al.*[22] Moreover, the applications of DES seem to be related to details shown in Figure 8, which indicates a wide variety of pathways for these solvents. Jessop[2] found similar results evaluating ionic liquids in comparison to conventional solvents, indicating that the novel solvent did not fill the entire range provided by conventional solvents, Figure 8. Nevertheless, it is worth highlighting that DES proved to be solvents with an extended range of polarity of common solvents and should be seen as complementary, and not just as a substitute, to conventional organic solvents.

CONCLUSIONS

In the study of green solvents, properties such as polarity, which can be described by the solvatochromic parameters, must be known. Despite these being widely available in conventional solvents, there is a lack of data for the nonconventional ones. This work proposes novel predictive models to estimate the K−T parameters for DES. After developing and testing an approach for organic solvents, based on COSMO-RS descriptors, this approach was successfully extended to DES. The obtained results for the DES training data set show R^2 values equal to or higher than 0.71. The overall performance of the developed model in the testing data set was outstanding, with a high R^2 (>0.86), showing a low error for the three K−T parameters. Moreover, a comparison between DES and organic solvents polarity revealed that DES can enhance the range of polarities, reinforcing their designer solvent character.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssuschemeng.1c02621.

COSMO-RS parameters obtained at 298.2 K for 175 conventional solvents and their solvatochromic parameters from Marcus database; solvatochromic parameters for different DES; COSMO-RS parameters obtained at 298.2 K for 44 DES; correlation models between K−T parameters and COSMO-RS descriptors for organic solvents; experimental and predicted solvatochromic parameters (α, β, and π^*) for different organic solvents; correlation models between K−T parameters and COSMO-RS descriptors for acid DES; correlation models between K−T parameters and COSMO-RS descriptors for nonacid DES; experimental and predicted solvatochromic parameters (α, β, and π^*) for DES obtained by this and other works; and figure with protic conventional solvents plotted according to their π^* and β values (PDF)

AUTHOR INFORMATION

Corresponding Authors

Marcos R. Mafra − *Department of Chemical Engineering, Federal University of Paraná (UFPR), Curitiba 81531-990, Brazil;* orcid.org/0000-0002-0018-6867; Phone: +55 3361-3586; Email: marcos.mafra@ufpr.br

João A.P. Coutinho − *CICECO—Aveiro Institute of Materials, Department of Chemistry, University of Aveiro (UA), Aveiro 3810-193, Portugal;* orcid.org/0000-0002-3841-743X; Phone: +351 234 370 200; Email: jcoutinho@ua.pt

Authors

José Pedro Wojeicchowski − *Department of Chemical Engineering, Federal University of Paraná (UFPR), Curitiba 81531-990, Brazil; CICECO—Aveiro Institute of Materials, Department of Chemistry, University of Aveiro (UA), Aveiro 3810-193, Portugal;* orcid.org/0000-0002-6790-8510

Dinis O. Abranches − *CICECO—Aveiro Institute of Materials, Department of Chemistry, University of Aveiro (UA), Aveiro 3810-193, Portugal;* orcid.org/0000-0003-0097-2072

Ana M. Ferreira − *CICECO—Aveiro Institute of Materials, Department of Chemistry, University of Aveiro (UA), Aveiro 3810-193, Portugal;* orcid.org/0000-0003-3057-5019

Complete contact information is available at: https://pubs.acs.org/10.1021/acssuschemeng.1c02621

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was developed within the scope of the project CICECO-Aveiro Institute of Materials, UIDB/50011/2020 and UIDP/50011/2020, financed by national funds through the Portuguese Foundation for Science and Technology/MCTES. J.P.W. is grateful for the scholarship (88881.361904/

2019-01) provided by CAPES (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Brazil) and by Banco Santander S.A (Brazil). M.R.M. is grateful to the Brazilian National Council for Scientific and Technological Development (CNPq—grant 308517/2018-0).

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